

Doom PWAD QA Report

Run 2b598745-6522-4b99-87ed-7ffced3eb74b | Map MAP01 | Outcome stuck

Executive Summary

Document the automated PWAD QA run, combine static map analysis with live play telemetry, and give map authors concrete evidence about completion, traversal, combat, and defects. The selected-difficulty gameplay evidence uses 13 spawned enemy/enemies. Raw static analysis found 14 enemies, but 1 are disabled at difficulty 1 by skill flags.

Pass/fail: map_navigation: FAIL combat_engagement: FAIL resource_balance: FAIL secret_coverage: FAIL

overall_verdict: FAIL

- The run ended with outcome 'stuck' after visiting 6 coarse position cluster(s).
- 0 kill(s) recorded against 13 enemy/enemies that spawn at difficulty 1.
- Selected-difficulty health_ratio=0.0844, ammo_ratio=0.1200.
- 0 secret(s) found.

The highest-severity open finding is 'Run stalled after repeated stuck decisions' at severity 1. Review this before publishing the map.

Test Item And Environment

Test item: MAP01 from WAD 76baf049-f52f-426b-934c-2b9a427e949b. Static analysis context: 43 things, 14 enemies, 12 items, 1 keys, 1 secret sectors, 174 linedefs, 45 sectors, estimated difficulty hard. At difficulty 1, 13 of 14 raw enemies spawn.

The run used IWAD freedoom2, difficulty 1, model gemini-3.1-flash-lite, and a maximum budget of 900 game ticks.

Metric	Value	Metric	Value
Status	completed	Duration	22
Actions	19	LLM calls	19
Final HP	101	Kills	0
Position samples	326	Movement units	2249.0
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/2b598745-6522-4b99-87ed-7ffced3eb74b.mp4		

Recording And Agent Quality

Recording	Value	Agent progress	Value
Status	ok	Progress score	10
Frames	132 total / 132 unique	Meaningful progress	4
Video shape	640 x 480 @ 15.0 FPS	Completed objects	4
Gameplay time	8.771s, 307 tic(s)	Quality flags	warning

Quality warnings: The requested action repeats a recent no-progress signature (take_action), so I am switching tactics.

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Static Analysis Context

Metric	Raw WAD	Difficulty 1	Notes
Enemies	14	13	1 hidden by skill/multiplayer flags
Items	12	12	0 hidden by skill/multiplayer flags
Secrets	1	1	Sector special 9
Geometry	174 linedefs	45 sectors	1536 x 872 units

Selected-difficulty estimate: hard. Health ratio: 0.0844. Ammo ratio: 0.12.

Coverage Evaluation

The automated playthrough recorded 19 gameplay event(s), 19 lockstep LLM/MCP decision(s), 326 movement samples, 6 position cluster(s), and 2249.0 map units of movement. The run ended with outcome 'stuck'. Combat coverage is evaluated against 13 enemy/enemies that spawn at difficulty 1, not against hidden raw THINGS. Recording quality status: ok; 132 frames, 132 unique frames, 8.8 seconds at 15.0 FPS. Automated playthrough progress score: 10 with 4 meaningful progress event(s). Quality flags: warning.

Objectives Covered

- **Map launch and initial state capture:** Run status: completed.
- **Director decision logging:** 19 decision event(s) recorded.
- **Movement telemetry:** 326 position sample(s) across 6 cluster(s).
- **Gameplay recording:** /media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/2b598745-6522-4b99-87ed-7ffcd3eb74b.mp4

Objectives Omitted

- **Verified map exit:** Run outcome was stuck.

Uncovered attributes: exit reliability, secret accessibility

Defects And Risks

3 defect(s) were detected. Primary risk: at least one detected defect needs author review.

Patterns: difficulty_spawn_mismatch: 1; softlock_navigation: 1; unreachable_secret: 1

Severity	Priority	Type	Tick	Description
1	1	softlock_navigation	271	Run stalled after repeated stuck decisions The run produced repeated stuck events with no meaningful progress. This usually indicates blocked navigation, unreachable objectives, or an automated control loop that should be reviewed with the recording and action trace.

3	2	difficulty_spawn_mismatch	0	Difficulty-specific thing flags hide map content At difficulty 1, 1 of 14 enemies from raw static analysis do not spawn in single-player because of Doom skill or multiplayer flags.
3	3	unreachable_secret	321	Secrets not reached Static analysis found secret sectors, but the automated playthrough did not enter one.

Risk Areas

- **Progression:** Map exit was not verified; outcome was stuck.
- **Difficulty flags:** 1 enemy/enemies are present in raw THINGS but do not spawn at difficulty 1.
- **softlock_navigation:** Run stalled after repeated stuck decisions
- **difficulty_spawn_mismatch:** Difficulty-specific thing flags hide map content
- **unreachable_secret:** Secrets not reached

Activity Log Summary

The pipeline statically analyzed the map, launched a ViZDoom MCP session, executed 19 lockstep MCP action(s), collected event and position telemetry, finalized recording artifacts, detected defects, and generated this report.

Run data is sparse or quality flags are present; treat gameplay conclusions as preliminary.

Limitations And Recommendations

1 raw enemy/enemies did not spawn at difficulty 1; combat conclusions only apply to spawned enemies. Automated play can miss secrets or optional combat paths that require map-specific intent. Single-run results should be confirmed with additional seeds or manual review before release decisions.

Good Quality Areas

- **Runtime stability:** The backend completed the run lifecycle.
- **Telemetry:** The run includes multiple position samples.
- **Evidence capture:** A gameplay recording artifact was produced.
- **Static context:** The report includes parsed map structure and thing counts.

Evidence Appendix

Notable Event Trace

Tick	Event	Position	HP	MCP Action	Stop reason
70	item_pickup	33.5, -171.2	101	move_to	arrived
271	stuck	2.5, -103.6	101	take_action	tics_complete
277	stuck	1.0, -125.3	101	take_action	tics_complete
280	stuck	0.2, -136.9	101	take_action	tics_complete
308	stuck	6.4, -56.8	101	retreat	complete
312	stuck	8.2, -32.1	101	take_action	tics_complete
318	stuck	11.1, 6.4	101	take_action	tics_complete
321	stuck	12.4, 23.3	101	take_action	tics_complete

LLM/MCP Decision Trace

#0 ticks 14 -> 27 tool move_to stop arrived Gemini API key is not configured; using deterministic fallback. Visible HealthBonus selected for collection.
#1 ticks 28 -> 44 tool move_to stop arrived Gemini API key is not configured; using deterministic fallback. Visible Clip selected for collection.
#2 ticks 45 -> 46 tool explore stop item_found Gemini API key is not configured; using deterministic fallback. No visible combat or pickup target is available, so a short bounded exploration step starts.
#3 ticks 47 -> 70 tool move_to stop arrived Gemini API key is not configured; using deterministic fallback. Visible ArmorBonus selected for collection.
#4 ticks 71 -> 72 tool explore stop item_found Gemini API key is not configured; using deterministic fallback. No visible combat or pickup target is available, so a short bounded exploration step starts.
#5 ticks 73 -> 95 tool move_to stop arrived Gemini API key is not configured; using deterministic fallback. Visible Shell selected for collection.
#6 ticks 96 -> 175 tool explore stop max_tics Gemini API key is not configured; using deterministic fallback. No visible combat or pickup target is available, so a short bounded exploration step starts.

#7 | ticks 176 -> 255 | tool explore | stop max_tics

Gemini API key is not configured; using deterministic fallback. No visible combat or pickup target is available, so a short bounded exploration step starts.

#8 | ticks 256 -> 258 | tool take_action | stop tics_complete

Gemini API key is not configured; using deterministic fallback. Recent exploration did not progress, so probing a nearby door/use interaction.

#9 | ticks 259 -> 261 | tool take_action | stop tics_complete

Gemini API key is not configured; using deterministic fallback. Recent exploration did not progress, so probing a nearby door/use interaction.

#10 | ticks 262 -> 264 | tool take_action | stop tics_complete

Gemini API key is not configured; using deterministic fallback. Recent exploration did not progress, so probing a nearby door/use interaction.

#11 | ticks 265 -> 267 | tool take_action | stop tics_complete

Gemini API key is not configured; using deterministic fallback. Recent exploration did not progress, so probing a nearby door/use interaction.

#12 | ticks 268 -> 271 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. A nearby door-like sector is known, so I am pressing USE and nudging forward.

#13 | ticks 272 -> 277 | tool take_action | stop tics_complete

Progress has not changed across repeated lockstep decisions, so I am forcing a bounded QA recovery probe. I am advancing straight under direct control instead of letting explore arc in place.

#14 | ticks 278 -> 280 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. I am probing for a switch or door interaction before declaring the area blocked.

#15 | ticks 281 -> 308 | tool retreat | stop complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. The direct probes did not progress yet, so I am retreating and rotating out of the loop.

#16 | ticks 309 -> 312 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. A nearby door-like sector is known, so I am pressing USE and nudging forward.

#17 | ticks 313 -> 318 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. I am advancing straight under direct control instead of letting explore arc in place.

#18 | ticks 319 -> 321 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (take_action), so I am switching tactics. I am probing for a switch or door interaction before declaring the area blocked.